8 **(a)** **(i)** **1.** mass B1

2. positive electric charge B1

(ii) magnetic force acts on moving charge in magnetic field, force on it may not be due to electric field B1

(b) **(i)** magnitude of magnetic flux density at wire *X*

$= \frac{\mu_0 I}{2\pi d} = \frac{4\pi\times10^{-7}\times290}{2\pi\times0.050}$ M1

$= 1.2\times10^{-3}$ T A0

(ii) $F = BIL \Rightarrow \frac{F}{L} = BI$

force per unit length
$= 1.2 \times 10^{-3} \times 290$ C1

$= 0.35$ N m^{-1} A1

(c) **(i)** electron moving perpendicular in magnetic field experience a magnetic force B1

force
$= Bqv = 4.64\times10^{-3}\times1.6\times10^{-19}\times2.9\times10^{7}$ M1

$= 2.15\times10^{-14} = 2.2\times10^{-14}$ N A0

(ii) magnetic field between the wires is not uniform B1

magnetic force / resultant force on electron not constant M1

so, claim is incorrect A0

(iii) curve downwards B1

(d) [Method 1]
electric force on electron $= qE = 1.6\times10^{-19}\times\frac{13500}{0.10} = 2.16\times10^{-14}$ N M1

either force equal to **(c)(i)**, so constant velocity
or force ≠ **(c)(i)**, so cannot be constant velocity A1

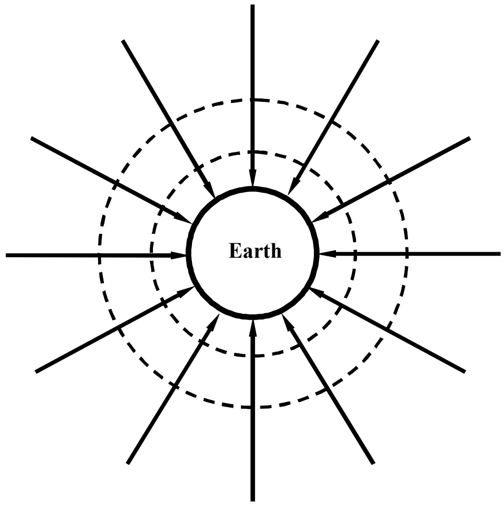
[Method 2]
For electron to move at constant velocity, magnetic force = electric force
$Bev = eE \Rightarrow v = \frac{13500}{0.10}/4.64\times10^{-3} = 2.91\times10^{7}$ m s^{-1} M1

either given speed = speed required, so constant velocity
or given speed < speed required, so cannot be constant velocity A1

(e) minimum **four** field lines directed towards Earth radially B1

minimum **two** concentric circles for equipotential surface B1

minimum **three** circles with increasing distance between one circle and next outer circle B1



(f) **(i)** gravitational field strength by earth = gravitational field strength by moon

$\frac{GM_{\text{Earth}}}{x^2} = \frac{GM_{\text{moon}}}{(3.84\times10^8 - x)^2}$ C1

$\frac{5.98\times10^{24}}{x^2} = \frac{7.35\times10^{22}}{(3.84\times10^8 - x)^2}$ C1

$x = 3.5 \times 10^8$ m A0

(ii) *either* acceleration (or resultant force) of spaceship and its displacement from P is in the same direction

or spaceship accelerates towards Earth if displaced left of P or towards moon if displaced right of P M1

so claim incorrect A1